



Homework 2

dplyr & ggplot2 with scales, guides, and themes

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Directions

- Submit **one Quarto document** (`.qmd`) to Brightspace:
 - `dan1-310-hw2-LASTNAME-FIRSTNAME.qmd`
(e.g., `dan1-310-hw2-choe-byeonghak.qmd`)
- **Due:** March 2, 2026, 2:00 P.M. (ET)
- For **visualization questions**, you must provide:
 1. the **ggplot2 code**, and
 2. a **written comment** (2–4 sentences) interpreting the corresponding figure.
- Unless a question says otherwise, use **dplyr** verbs (`filter()` , `distinct()` , `select()` , `mutate()` , `group_by()` , `summarise()` , `arrange()` , `count()` , etc.) and **ggplot2**.



Setup

```
library(tidyverse)
library(ggthemes)
library(hrbrthemes)
library(skimr)
```

Part 1. Data Visualization

Replicate given ggplot figures.

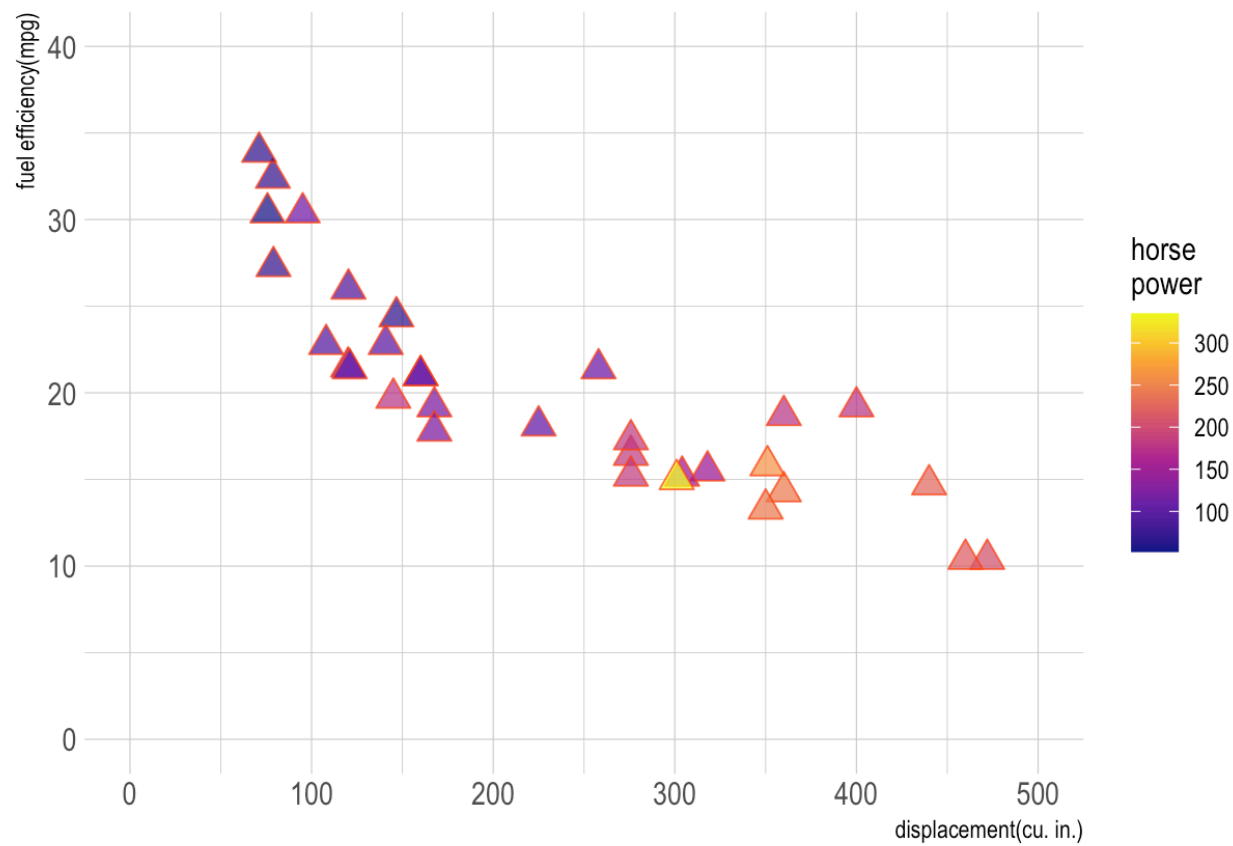
Question 1

Use `datasets::mtcars` for Question 1.

	mpg	cyl	disp	hp	drat	wt	qsec	vs
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1

1-10 of 32 rows | 1-9 of 12 columns

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```
# YOUR CODE HERE
```

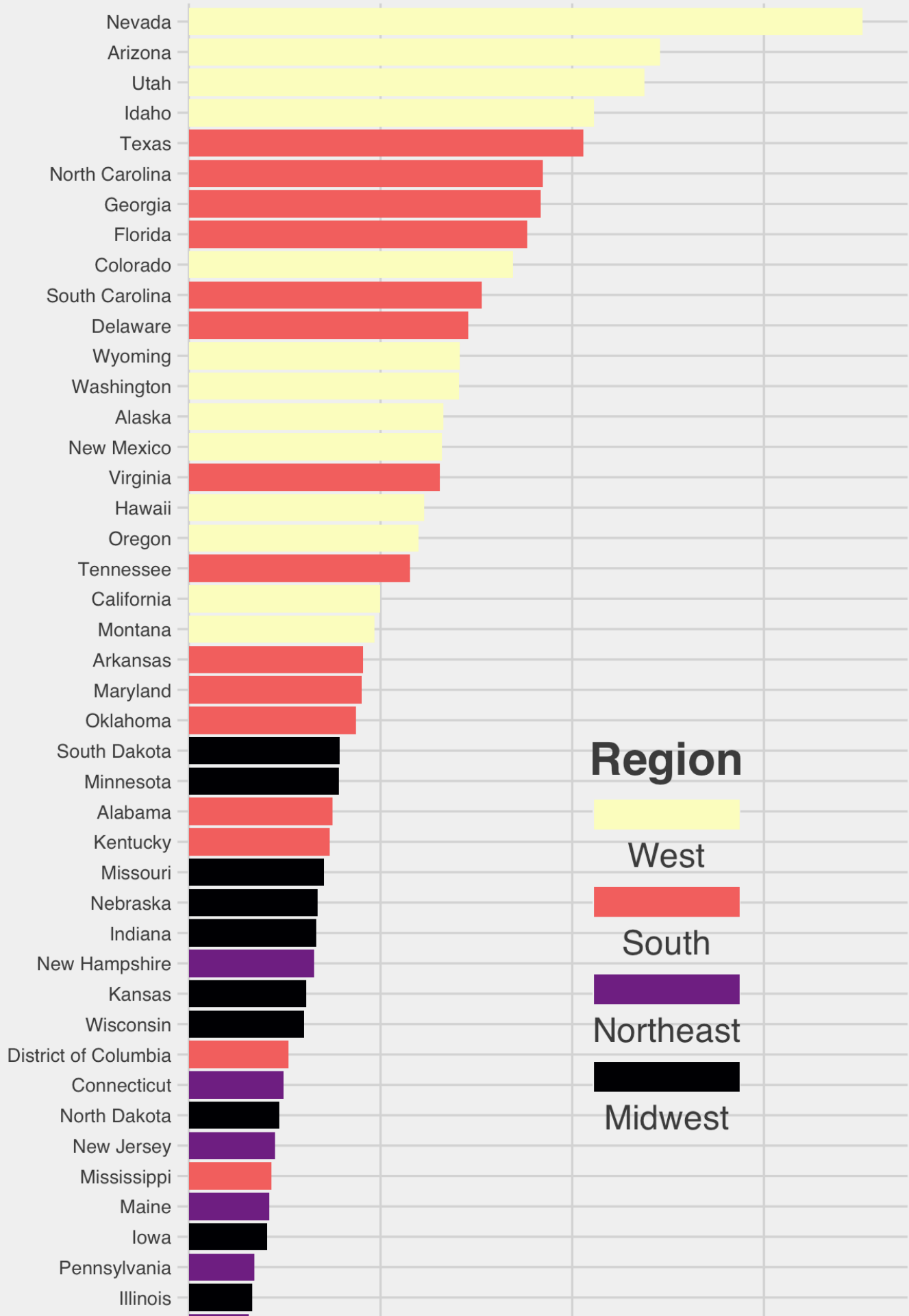
Question 2

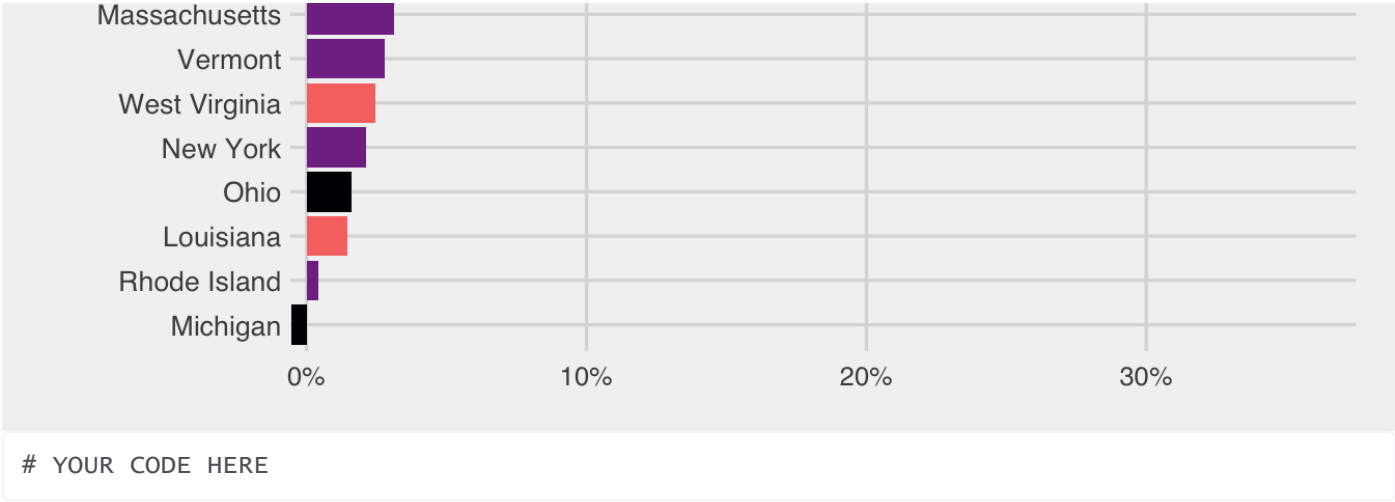
Use the following data.frame for Question 2.

```
popgrowth_df <- read_csv(  
  'https://bcdanl.github.io/data/popgrowth.csv')  
)
```

region	division	state	pop2000
<chr>	<chr>	<chr>	<dbl>
Midwest	East North Central	Michigan	9938444
Northeast	New England	Rhode Island	1048319
South	West South Central	Louisiana	4468976
Midwest	East North Central	Ohio	11353140
Northeast	Middle Atlantic	New York	18976457
South	South Atlantic	West Virginia	1808344

region	division	state	pop2000
<chr>	<chr>	<chr>	<dbl>
Northeast	New England	Vermont	608827
Northeast	New England	Massachusetts	6349097
Midwest	East North Central	Illinois	12419293
Northeast	Middle Atlantic	Pennsylvania	12281054
1-10 of 51 rows 1-4 of 7 columns			Previous 1 2 3 4 5 6 Next





Question 3

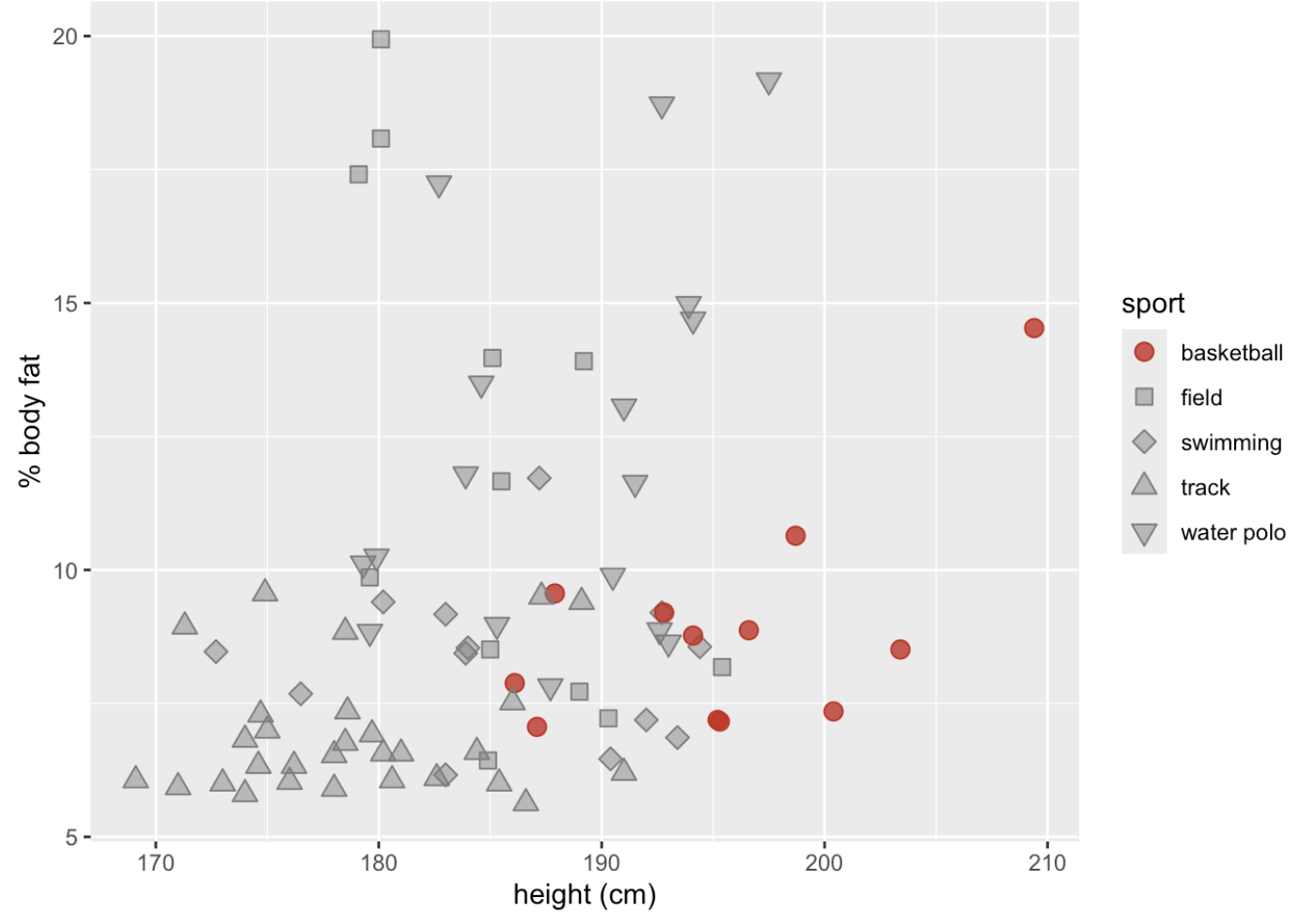
Use the following data.frame for Question 3.

```
male_Aus <- read_csv('https://bcdanl.github.io/data/aus_athletics_male.csv')
```

rcc	wcc	hc	hg	ferr	bmi	ssf	pcBfat	lbm	height
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
5.13	7.1	46.8	15.9	34	22.46	44.5	8.47	61	172.7
4.83	7.6	45.2	15.2	97	23.88	41.8	7.68	69	176.5
5.09	4.7	46.6	15.9	55	23.68	33.7	6.16	74	183.0
5.17	4.1	44.9	15.0	76	23.15	50.9	8.56	80	194.4
5.11	6.7	46.1	15.6	93	22.32	40.5	6.86	78	193.4
5.03	7.1	45.1	15.2	46	24.02	51.2	9.40	71	180.2
5.32	6.0	47.5	16.3	155	23.29	54.4	9.17	71	183.0
4.75	8.6	45.5	15.2	99	25.11	52.3	8.54	78	184.0
5.34	6.6	48.6	16.5	35	22.81	57.0	9.20	77	192.7
4.87	4.8	44.9	15.4	124	26.25	65.3	11.72	81	187.2

1-10 of 83 rows | 1-10 of 13 columns

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```
# YOUR CODE HERE
```

Question 4

Use the following data.frame for Question 4.

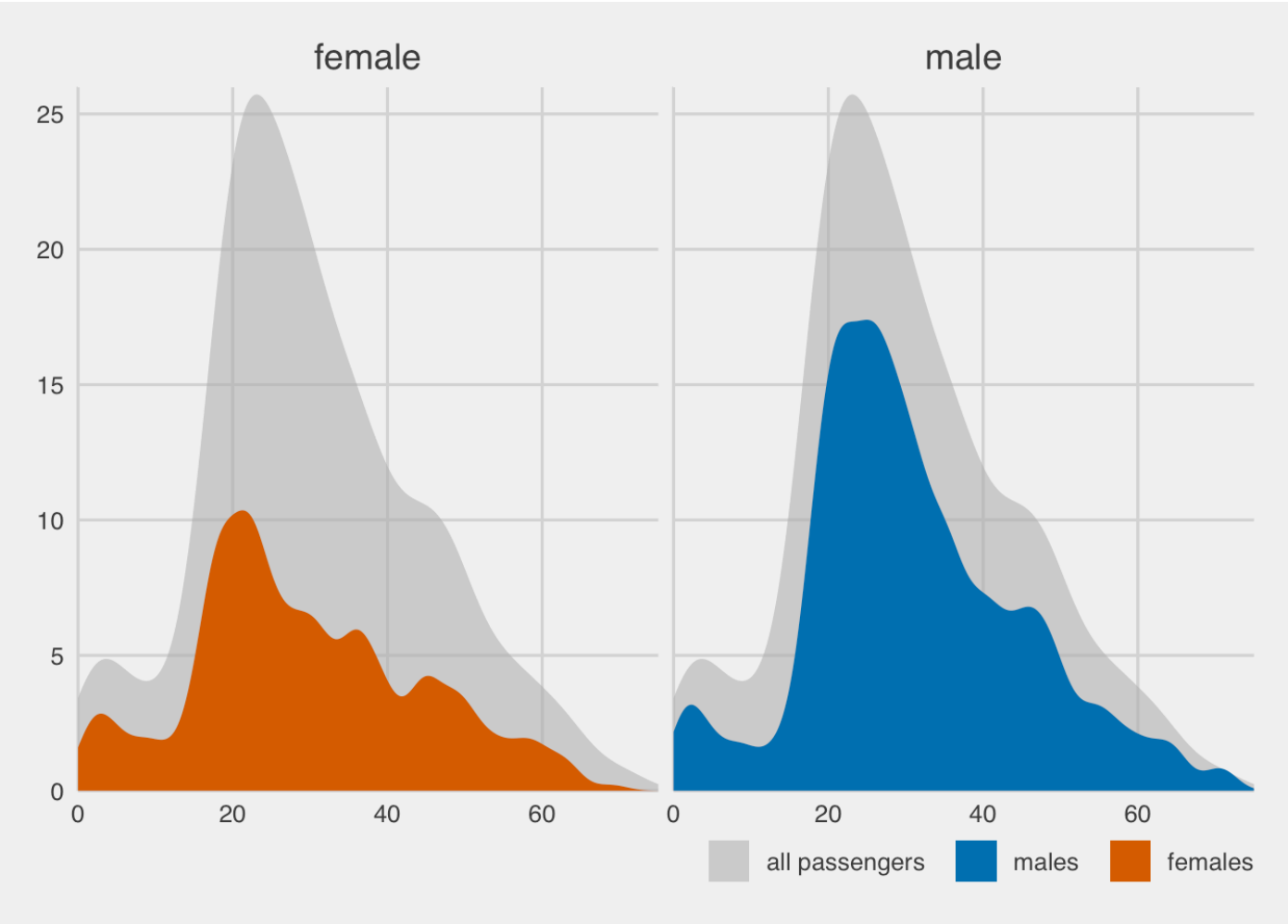
```
titanic <- read_csv('https://bcdanl.github.io/data/titanic_cleaned.csv')
```

class	age	gender	survived
<chr>	<dbl>	<chr>	<lgl>
1st	29.00	female	TRUE
1st	2.00	female	FALSE
1st	30.00	male	FALSE
1st	25.00	female	FALSE
1st	0.92	male	TRUE
1st	47.00	male	TRUE

class	age	gender	survived
<chr>	<dbl>	<chr>	<lgl>
1st	63.00	female	TRUE
1st	39.00	male	FALSE
1st	58.00	female	TRUE
1st	71.00	male	FALSE

1-10 of 756 rows

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YOUR CODE HERE

Question 5

Use the following data.frame for Question 5.

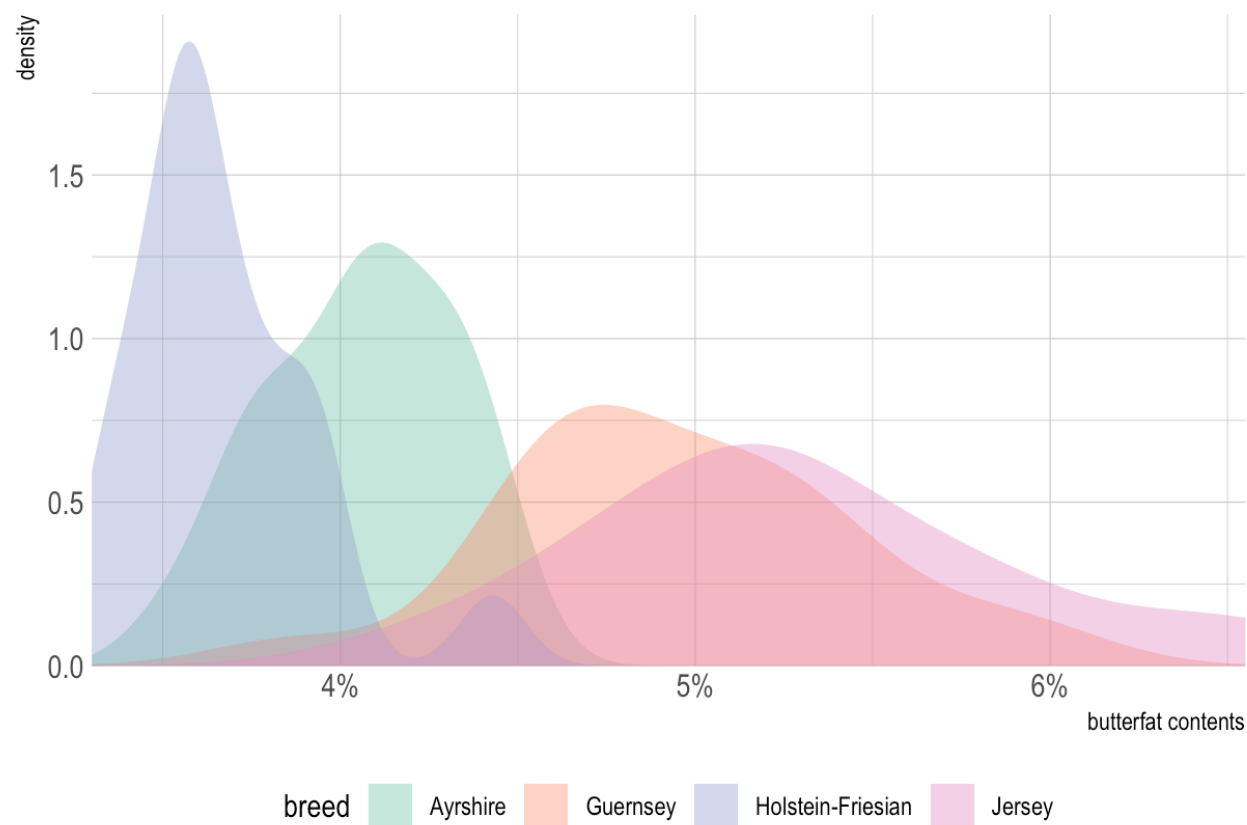

```
cows_filtered <- read_csv(  
  'https://bcdanl.github.io/data/cows_filtered.csv')  

```

butterfat	age	breed
<dbl>	<chr>	<chr>
3.74	Mature	Ayrshire
4.01	Mature	Ayrshire
3.77	Mature	Ayrshire
3.78	Mature	Ayrshire
4.10	Mature	Ayrshire
4.06	Mature	Ayrshire
4.27	Mature	Ayrshire
3.94	Mature	Ayrshire
4.11	Mature	Ayrshire
4.25	Mature	Ayrshire

1-10 of 80 rows

Previous123456...8Next



```
# YOUR CODE HERE
```

Part 2. Data Transformation + Visualization

Load the data.frame for Part 2.

```
path <- 'https://bcdanl.github.io/data/GHG_emissions_by.csv'
ghg_emissions <- read_csv(path)
```

Party	Year	Sector	
<chr>	<dbl>	<chr>	
Australia	1990	Energy	
Australia	1991	Energy	
Australia	1992	Energy	
Australia	1993	Energy	
Australia	1994	Energy	
Australia	1995	Energy	
Australia	1996	Energy	
Australia	1997	Energy	
Australia	1998	Energy	
Australia	1999	Energy	

1-10 of 7,224 rows | 1-3 of 4 columns

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Question 6

Provide both ggplot code and a simple comment to describe the yearly trend of GHG emissions for each sector.

```
# YOUR CODE HERE
```

Question 7

Provide both ggplot code and a simple comment to describe the yearly trend of United States of America's GHG emissions for each sector.

```
# YOUR CODE HERE
```

Question 8

For each party, calculate the yearly percentage change in GHG emissions for each sector.

```
# YOUR CODE HERE
```

Question 9

Which party has reduced GHG emissions most from 1990 level to 2017 level in terms of the percentage change in GHG emissions?

```
# YOUR CODE HERE
```

Question 10

Which sector has reduced GHG emissions most from 1990 level to 2017 level in terms of the percentage change in GHG emissions?

```
# YOUR CODE HERE
```



End of Homework 2

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